

Assignment 5 - Data Science

Insurance Policy Cost Prediction

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Abstract

In this analysis we try to study the factors that influence the policy cost determined by an insurance company. Some key findings of this analysis are: Yearly Earnings, Financial Rating, Prior Claims, and Wellness Index are the most influential factors in determination of Policy Cost.

1 Introduction

1.1 Background:

Correct determination of *Policy Cost* is a crucial part of the work that insurance companies do everyday; And machine learning is a powerful tool that can be used in order to have an accurate prediction on this objective. In this analysis we work with a dataset gathered from an insurance company, which in turn, includes many valuable features about **Demographic**, **Health**, and **Finance** of the person whom is been giving service to, as well as, information about the **Property/Asset** itself.

1.2 Objectives:

Below provides the flow of our analysis in a nutshell:

1. First we will try to gain a basic understanding about our data through **Exploratory Data Analysis** section.

2. Then we will apply a preprocessing step, to ensure the quality of the data.
3. Next, we shall apply different **Feature Engineering** techniques, in order to help our models, capture patterns from the data.
4. After that, we are going to train our models and evaluate them and see which one performs the best.
5. Then we shall extract the importances of the features, obtained by each of the models and try to provide useful insights for our business.

2 Data

2.1 An Overview:

The data has been gathered from a Kaggle competition. The dataset includes 1.1 million and 21 columns in total. The features include both categorical (both nominal and ordinal) and continuous values.

2.2 Exploratory Data Analysis

2.2.1 Missing Values:

The missing values for each of the columns are shown in the figure 1.

ID	0
Years Lived	17219
Sex	0
Yearly Earnings	41151
Relationship Status	16982
Dependent Count	100703
Academic Standing	0
Job Title	328542
Wellness Index	68046
Region	0
Coverage Class	0
Prior Claims	333860
Automobile Age	6
Financial Rating	126405
Coverage Period	1
Coverage Commencement	0
Client Review	71297
Tobacco Use	0
Physical Activity	0
Asset Category	0
Policy Cost	0

Figure 1: Missing Values

2.2.2 Uni-Variate Analysis:

We can see the distribution of all of the numerical features in the figure 2; As you can see, the distribution of *Yearly Earnings* is skewed to the right, but other features are distributed as normal or uniform.



Figure 2: Distribution of Numerical Features

2.2.3 Bi-Variate Analysis:

We can see the correlation matrix of all of the numerical features in the figure 3; We observe that no pair of the features have a high correlation with each other.



Figure 3: Heatmap of the Correlation Matrix

2.3 Data Preprocessing

2.3.1 Imputation of the Missing Values:

We will impute the missing values by following the rule below:

- **Datetime & Categorical Features** → by Mode
- **Continuous Features** → by Median

2.4 Feature Engineering

2.4.1 Feature Expansion:

We will expand our features by extracting the following features from *Coverage Commencement*:

- *CoverageCommencement_Year*
- *CoverageCommencement_Month*
- *CoverageCommencement_DayOfWeek*
- *CoverageCommencement_DayOfYear*
- *CoverageCommencement_Season*

2.4.2 Feature Transformation:

We shall transform the feature, *Yearly Earnings*, by **Square Root** function, in order to make it's distribution more normal.

2.4.3 Feature Encodings:

We will apply **Feature Encodings** as follows:

- **Label Encodings:**
 - *Academic Standings*
 - *Coverage Class*
 - *Client Review*
 - *Physical Activity*
- **Target Encodings (with cross validation):**
 - *Sex*
 - *Relationship Status*
 - *Job Title*
 - *Region*
 - *Tobacco Use*
 - *Asset Category*
 - *CoverageCommencement_DayOfWeek*
 - *CoverageCommencement_Season*

3 Models & Evaluation

3.1 Splitting:

We will use 80% of the data as train, and 20% as validation.

3.2 Subsampling:

We shall use 1% of the data to decide on the best hyperparameters. Also we shall re-train and evaluate the models with the best hyperparameters with only a 5% of the train and validation set.

3.3 Evaluation Metric:

We will use the metric **RMSLE** to evaluate our models. The formula is given below:

$$RMSLE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\log(1 + \hat{y}_i) - \log(1 + y_i))^2}$$

3.4 Models:

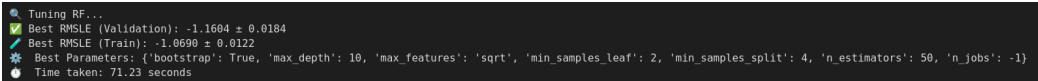
The models that we will use in this analysis are as follows:

1. **Random Forests**
2. **AdaBoost**
3. **XGBoost**
4. **Light Boost**
5. **Cat Boost**
6. **Polynomial Regression with Ridge Regularization**

3.5 Hyperparameter Tuning:

The results of **Hyperparameter Tuning** for each of the models is as follows:

3.5.1 Random Forests



```
Tuning RF...
✓ Best RMSLE (Validation): -1.1604 ± 0.0184
✓ Best RMSLE (Train): -1.0690 ± 0.0122
✱ Best Parameters: {'bootstrap': True, 'max_depth': 10, 'max_features': 'sqrt', 'min_samples_leaf': 2, 'min_samples_split': 4, 'n_estimators': 50, 'n_jobs': -1}
● Time taken: 71.23 seconds
```

Figure 4: Hyperparameter Tuning Results for Random Forests

3.5.2 AdaBoost

```
Tuning AdaBoost...
✓ Best RMSLE (Validation): -1.2231 ± 0.0171
✓ Best RMSLE (Train): -1.2195 ± 0.0123
✎ Best Parameters: {'learning_rate': 0.1, 'n_estimators': 50}
⌚ Time taken: 12.73 seconds
```

Figure 5: Hyperparameter Tuning Results for AdaBoost

3.5.3 XGBoost

```
Tuning XGBoost...
✓ Best RMSLE (Validation): -1.1566 ± 0.0178
✓ Best RMSLE (Train): -1.0877 ± 0.0139
✎ Best Parameters: {'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 50, 'subsample': 0.8}
⌚ Time taken: 7.34 seconds
```

Figure 6: Hyperparameter Tuning Results for XGBoost

3.5.4 Light Gradient Boost

```
Tuning LightGBM...
✓ Best RMSLE (Validation): -1.1551 ± 0.0151
✓ Best RMSLE (Train): -1.0730 ± 0.0141
✎ Best Parameters: {'colsample_bytree': 0.8, 'learning_rate': 0.05, 'max_depth': 10, 'min_child_samples': 10, 'n_estimators': 100, 'subsample': 0.8}
⌚ Time taken: 20.11 seconds
```

Figure 7: Hyperparameter Tuning Results for Light Gradient Boost

3.5.5 Cat Boost

```
Tuning CatBoost...
✓ Best RMSLE (Validation): -1.1590 ± 0.0190
✓ Best RMSLE (Train): -1.1027 ± 0.0127
✎ Best Parameters: {'depth': 6, 'iterations': 200, 'learning_rate': 0.05}
⌚ Time taken: 13.76 seconds
```

Figure 8: Hyperparameter Tuning Results for Cat Boost

3.5.6 Polynomial Regression with Ridge Regularization

```
Tuning PolynomialRidge...
✓ Best RMSLE (Validation): -1.1655 ± 0.0206
✓ Best RMSLE (Train): -1.1361 ± 0.0148
✎ Best Parameters: {'poly_degree': 2, 'reg__alpha': 10.0, 'reg__fit_intercept': False}
⌚ Time taken: 37.77 seconds
```

Figure 9: Hyperparameter Tuning Results for Polynomial Regression with Ridge Regularization

3.6 Re-Training & Evaluation

The results obtained after re-training and then evaluation can be seen in the table 1.

Model	Train RMSLE	Validation RMSLE
AdaBoost	1.2418	1.2315
CatBoost	1.1518	1.1499
LightGBM	1.1385	1.1416
PolynomialRidge	1.1636	1.1561
RF	1.1365	1.1543
XGBoost	1.1459	1.1485

Table 1: RMSLE scores for each of the models on training and validation sets after re-training.

3.7 Feature Importances

As we can see in the table 2, *Yearly Earnings*, *Financial Rating*, *Prior Claims*, *CoverageCommencement_DayOfYear* and *Wellness Index* are among the most 10 influential features, across most of the models. We will provide insights about how these feature importances are going to help our business in the next section.

Table 2: Top 10 Feature Importances per Model

Model	Feature	Importance
CatBoost (Impurity)	Yearly Earnings	23.3703
	Financial Rating	13.9649
	Prior Claims	12.2697
	Wellness Index	7.4267
	CoverageCommencement_Year	7.3464
	Automobile Age	3.1645
	CoverageCommencement_DayOfWeek	2.7919
	Years Lived	2.6373
	Client Review	2.5793
	Physical Activity	2.2474
XGBoost (Impurity)	CoverageCommencement_Year	0.1165
	Prior Claims	0.0833
	Yearly Earnings	0.0640
	Financial Rating	0.0623
	Wellness Index	0.0484
	Physical Activity	0.0450
	Client Review	0.0435
	CoverageCommencement_DayOfWeek	0.0395
	Coverage Period	0.0389
	CoverageCommencement_DayOfYear	0.0378
Polynomial Ridge (Permutation)	CoverageCommencement_Month	0.0998
	CoverageCommencement_DayOfYear	0.0877
	Yearly Earnings	0.0103
	Financial Rating	0.0085
	Prior Claims	0.0081
	CoverageCommencement_Year	0.0055
	Dependent Count	0.0019
	CoverageCommencement_Season	0.0018
	Sex	0.0016
	Tobacco Use	0.0016
Random Forest (Impurity)	Yearly Earnings	0.1679
	Financial Rating	0.1149
	Wellness Index	0.0993
	CoverageCommencement_DayOfYear	0.0831
	Years Lived	0.0657
	Automobile Age	0.0553
	Prior Claims	0.0451
	Coverage Period	0.0405
	CoverageCommencement_DayOfWeek	0.0347
	CoverageCommencement_Year	0.0339
LightGBM (Impurity)	Yearly Earnings	463.0000
	Wellness Index	404.0000
	Financial Rating	373.0000
	CoverageCommencement_DayOfYear	301.0000
	Years Lived	215.0000
	Automobile Age	136.0000
	CoverageCommencement_Year	122.0000
	Prior Claims	120.0000
	Coverage Period	116.0000
	CoverageCommencement_DayOfWeek	103.0000
AdaBoost (Impurity)	Yearly Earnings	0.4293
	Financial Rating	0.2285
	Prior Claims	0.1215
	Wellness Index	0.0608
	CoverageCommencement_Year	0.0517
	Years Lived	0.0363
	CoverageCommencement_DayOfYear	0.0214
	Coverage Period	0.0129
	Dependent Count	0.0071
	Academic Standing	0.0063

4 Conclusion & Recommendation

Based on the previous analysis in the **Feature Importances** section, we will provide insights about how these information about the feature could help our business:

- We could provide user-segmented discounts on the premium services based on their *Finance* information such as *Yearly Earnings* and *Financial Rating*.
- It's possible to penalize the users, who have a high prior, based on the time that it has passed from their coverage commencement.
- We can provide an special discount to those who have a decent *Wellness Index*.